OHIO UNIVERSITY

Guest Lecture at Ohio University, Oct 30th, 2025

Generative AI for Healthcare Modeling

Ziyang Song

Assistant Professor, EECS, Ohio University

Ph.D., CS, McGill University

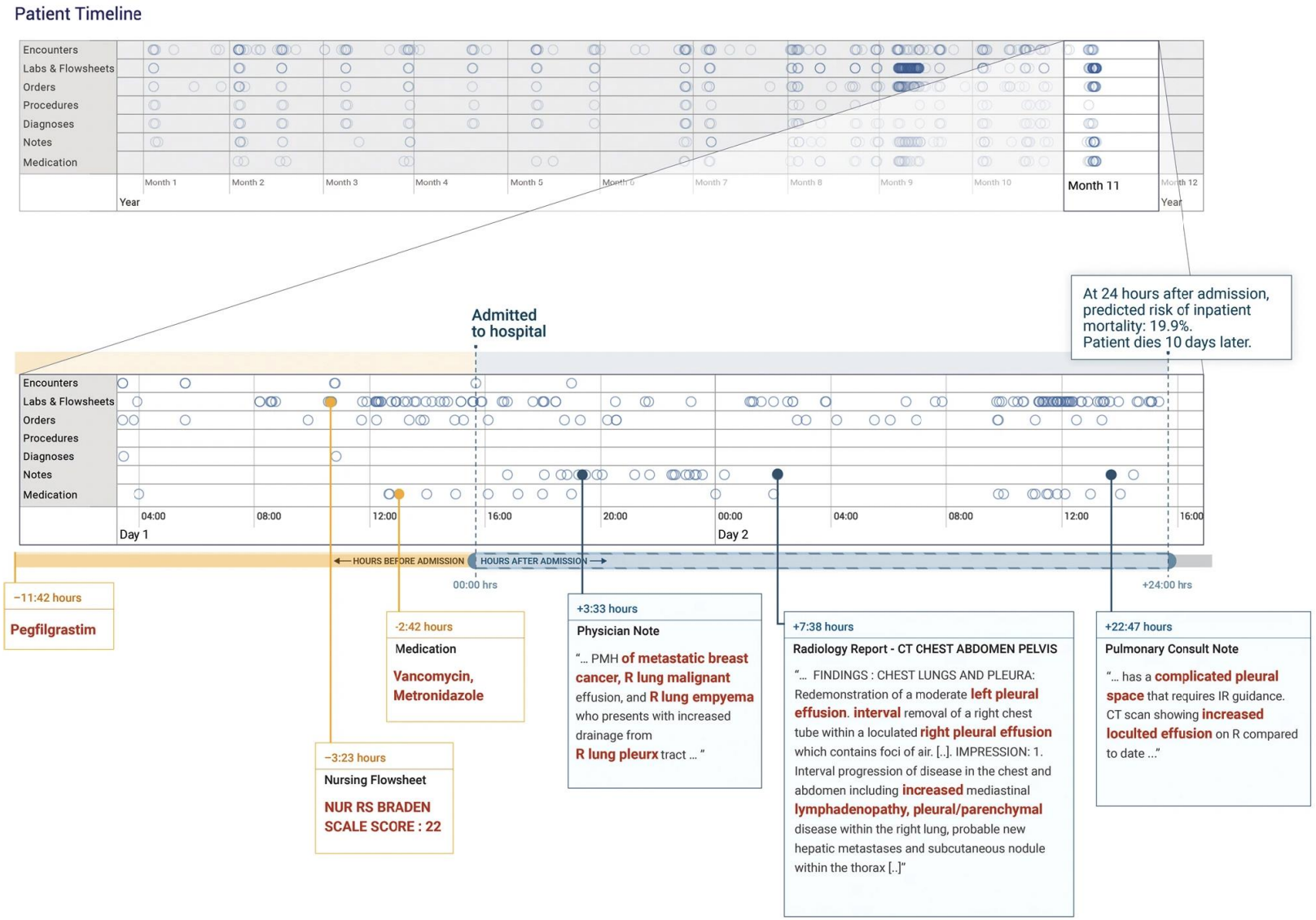
Research focus: AI for health

Outline

- Background
- GPT and Recurrent Transformer
- TrajGPT for Trajectory Modeling
- Representation Learning for Medical Concepts
- Reinforcement Learning for LLMs

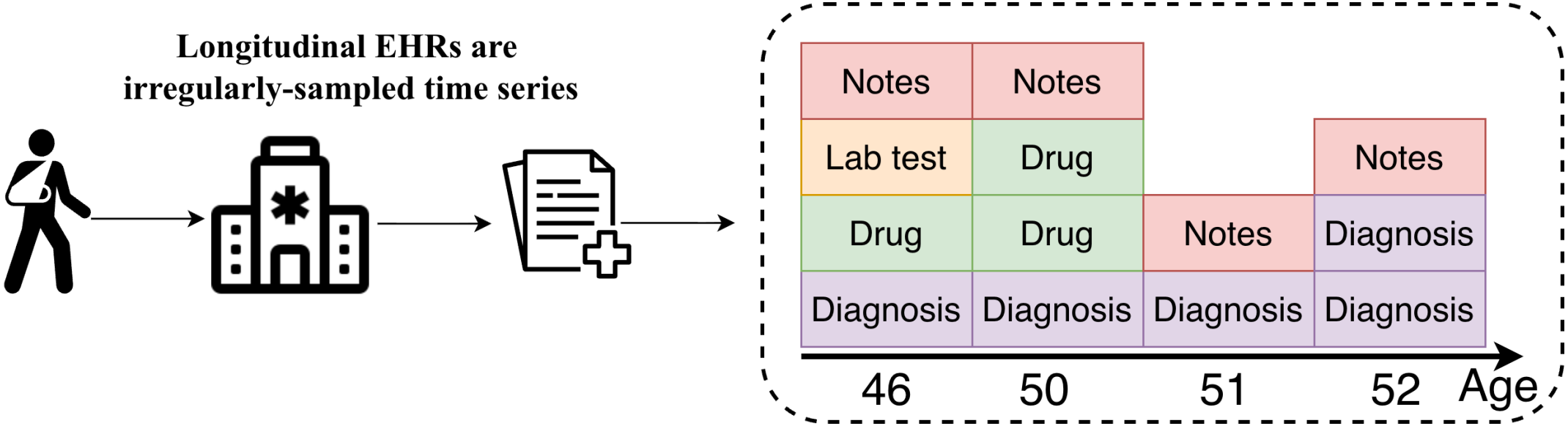
Medical Sequential Modeling

- Electronic Health Records (EHRs) contain rich longitudinal information



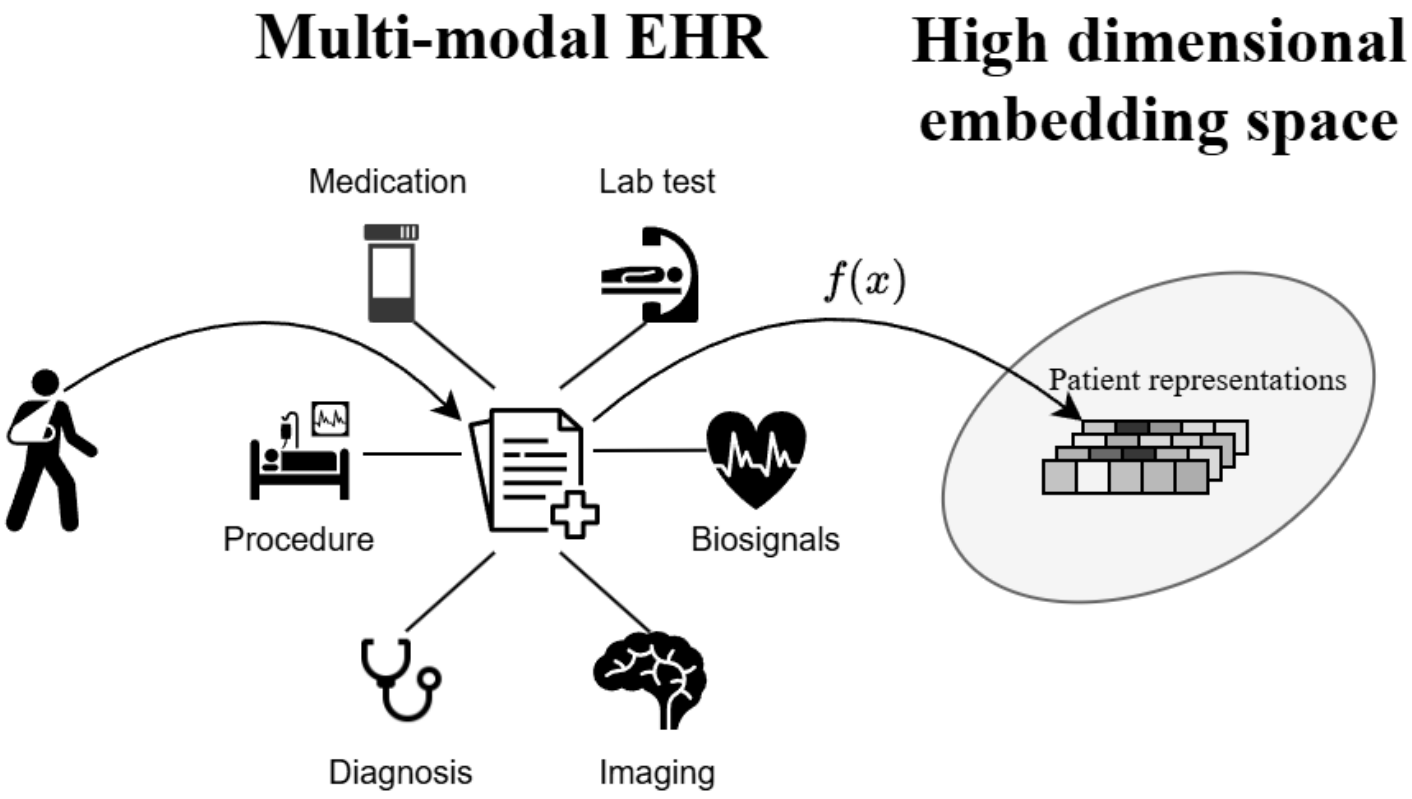
Irregularly-sampled Event Sequences

- Time-series can be divided into two categories:
  - Continuous time-series, like biosignals
    - Consecutive observations: $[x_{t=0}, x_{t=1}, \dots, x_{T-1}, x_T]$
  - Irregularly-sampled time-series, like longitudinal EHR data
    - Irregular observations with different interval: $[x_{t=20}, x_{t=25}, \dots, x_{42}, x_{43}]$
- Each patient medical records can be viewed as a irregularly-sampled event sequence for sequence modeling



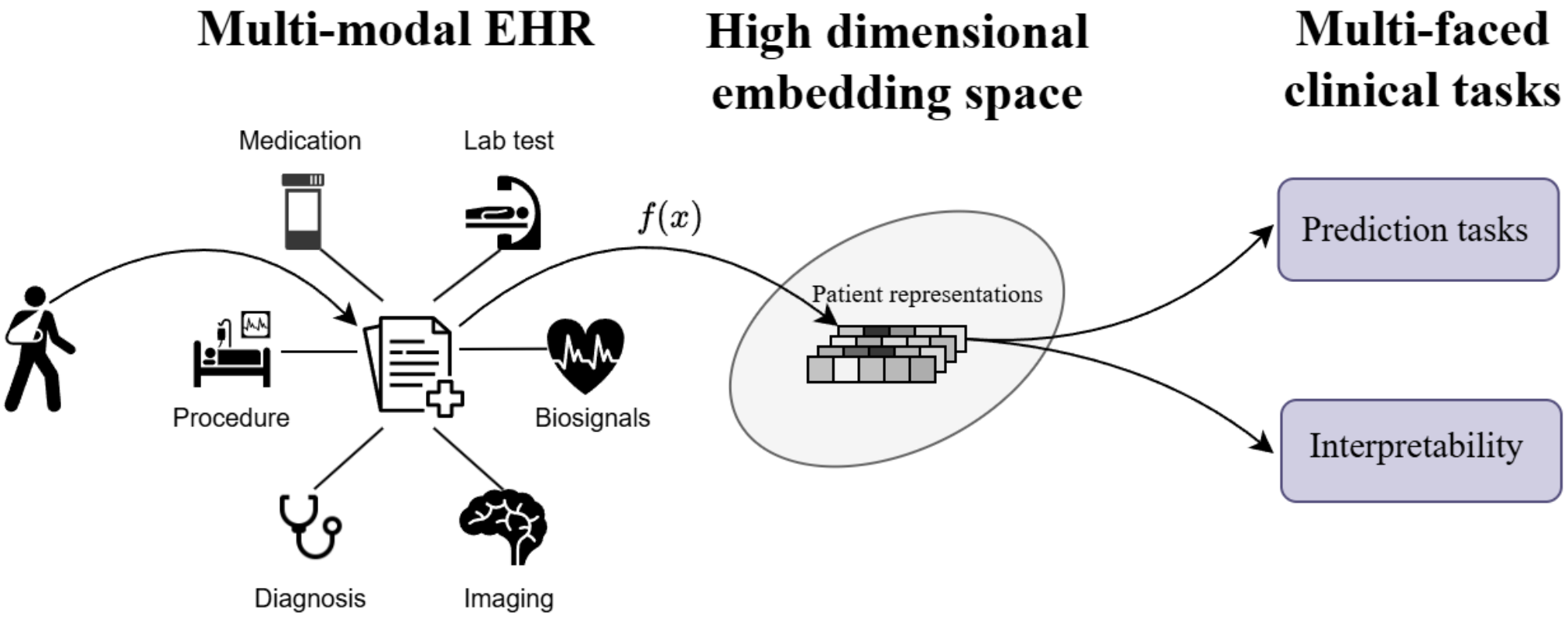
Representation Learning for Complex EHR Data

- Project a patient's multi-modal EHR data into a high-dimensional embedding



Representation Learning for Prediction

- Project a patient's multi-modal EHR data into a high-dimensional embedding
- Patient representations enable prediction tasks with interpretability

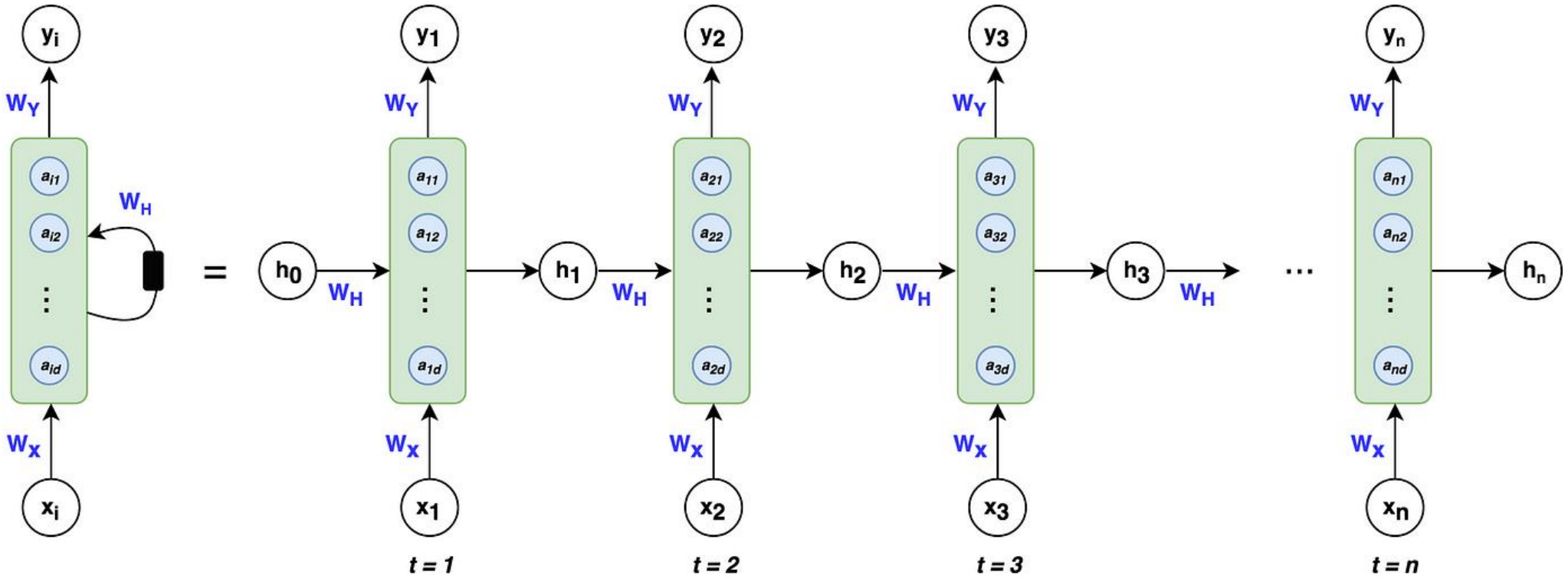


Outline

- Background
- GPT and Recurrent Transformer
- TrajGPT for Trajectory Modeling
- Representation Learning for Medical Concepts
- Reinforcement Learning for LLMs

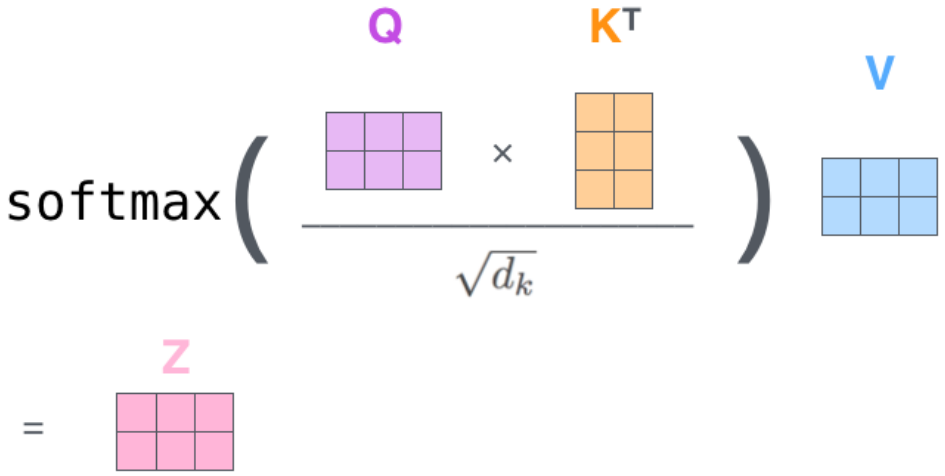
RNN

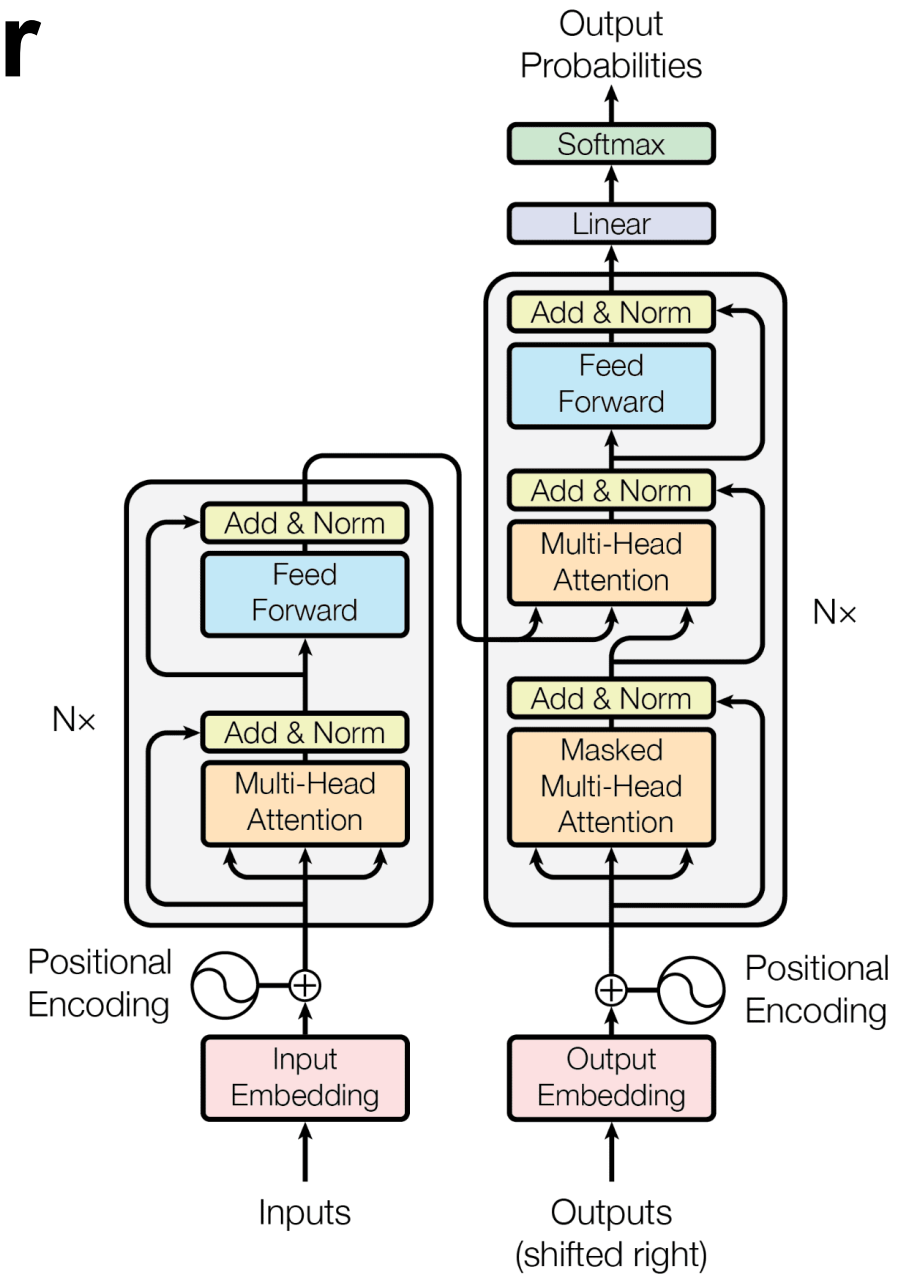
- RNNs are designed to model sequential data:
  - Training cannot be parallelized
  - Recurrent inference is fast
- Fail to scale-up by stacking layers and increasing hidden dimension



Transformer

- Transformer is an encoder-decoder architecture with a self attention mechanism and positional encoding
- The computational bottleneck is attention's quadratic complexity

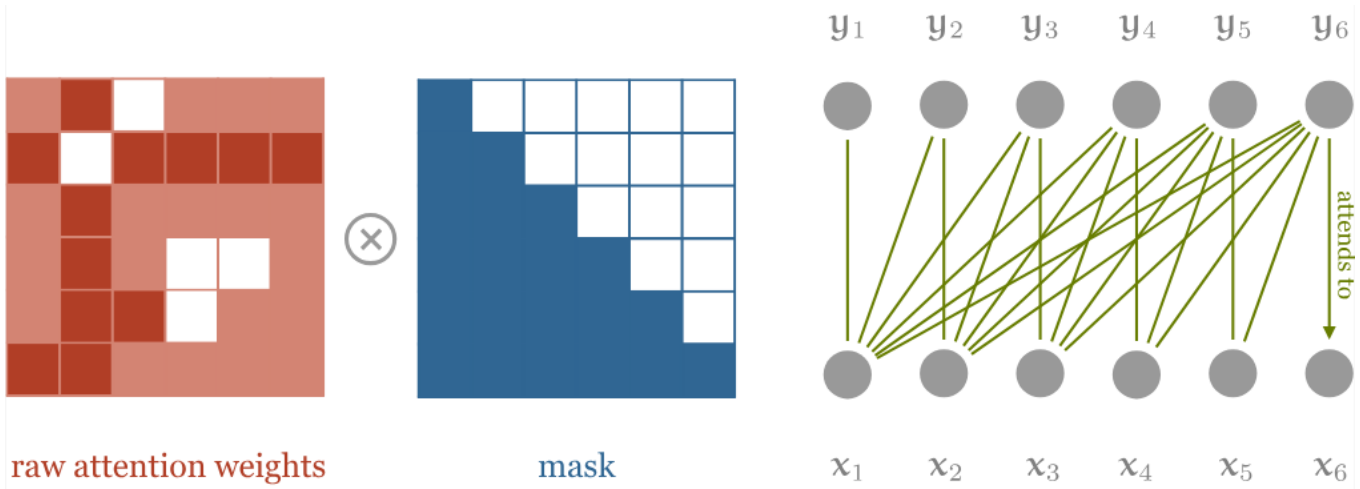


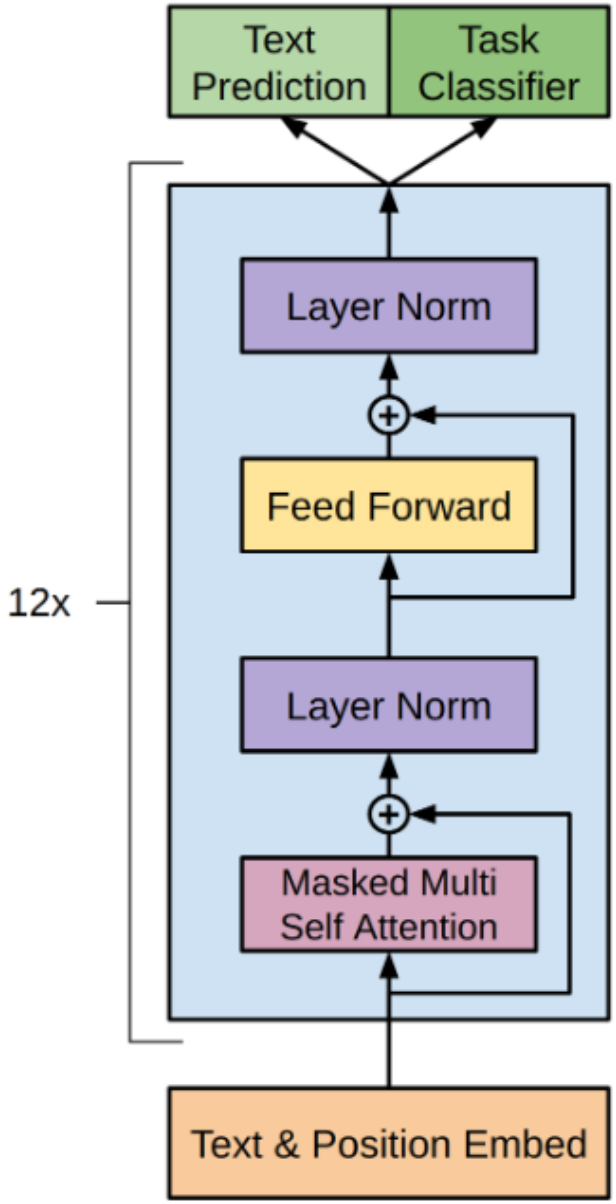


GPT

- GPT is a decoder-only architecture with a casual attention mechanism
- For each token, casual attention only takes the past tokens using a casual mask:

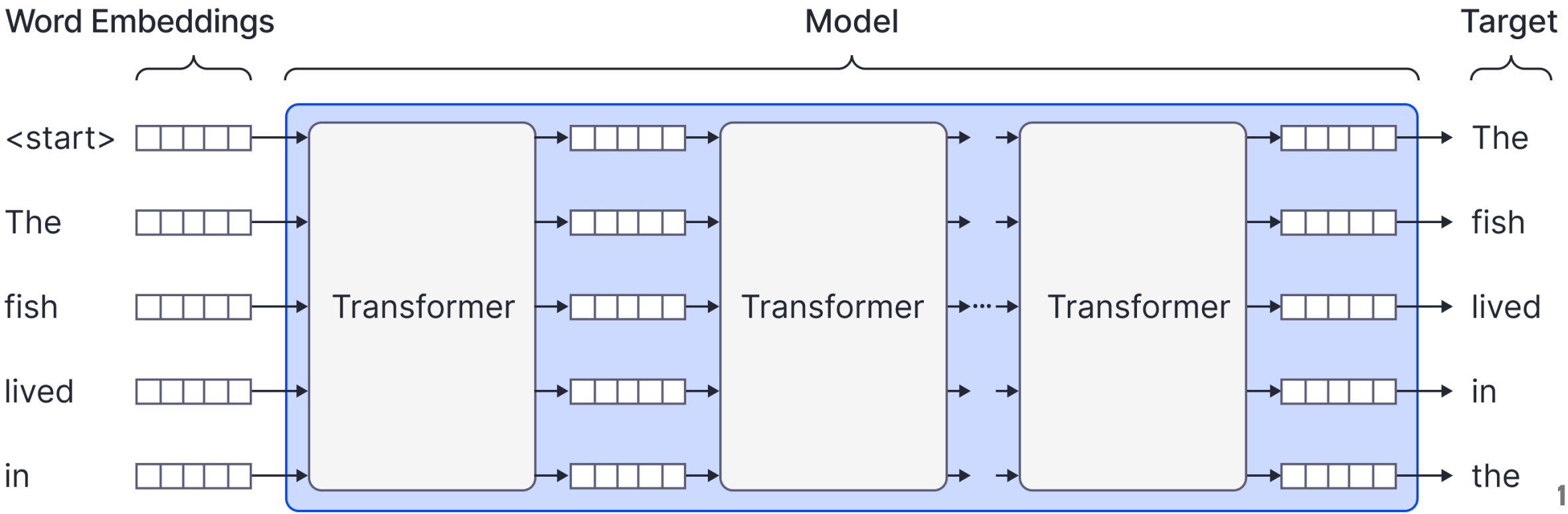
$$O = \mathrm{softmax}(QK^{\top} \odot M)V$$





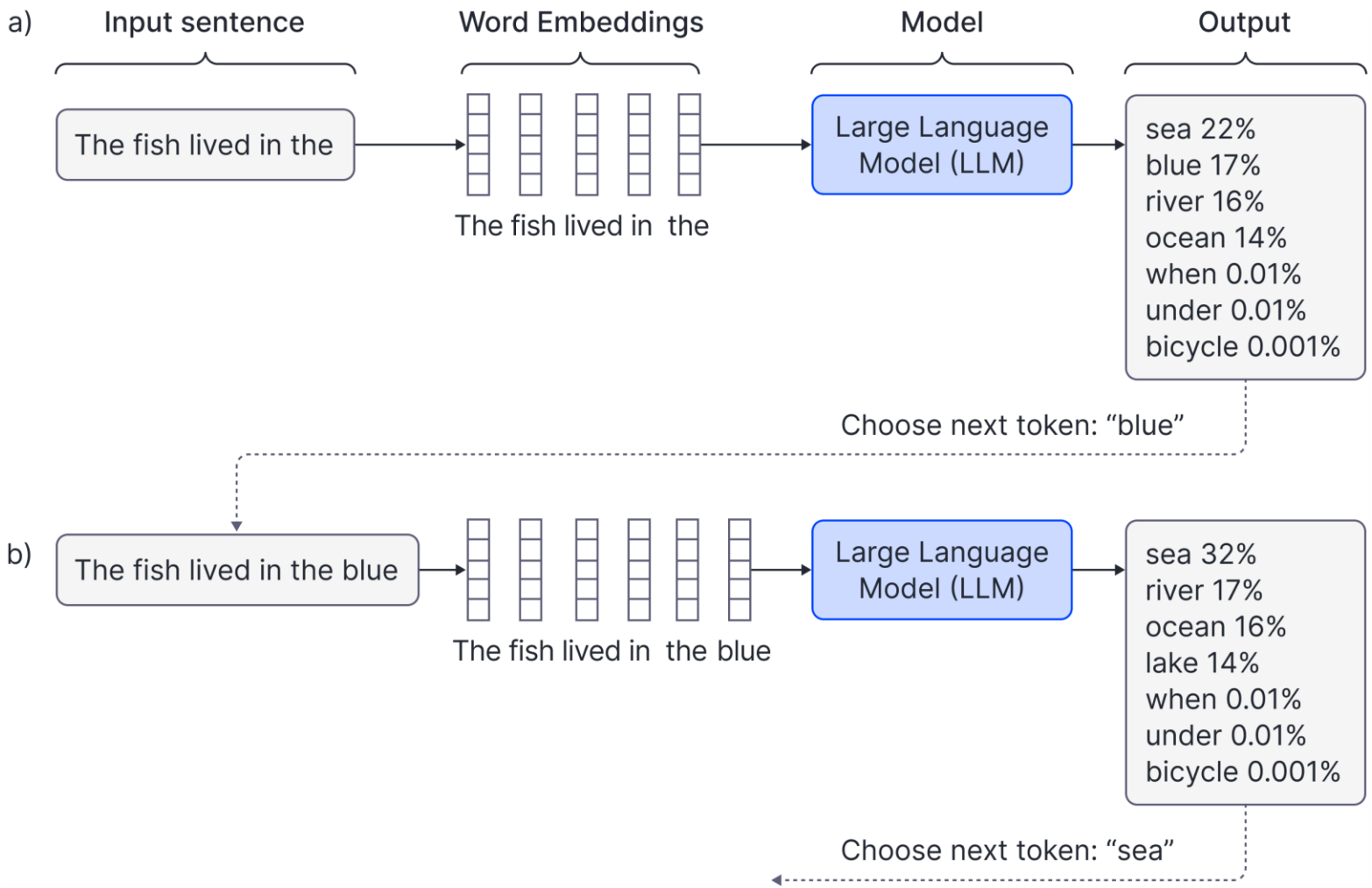
Next-token Prediction

- GPT uses a next-token prediction task for pre-training, i.e., predicting the next token using all previous tokens
- This pre-training task will learn contextualized representation for downstream tasks, such as sequence generation



Sequence Generation

- For each prediction, GPT generates a token and add it to existing sequence
- GPT architecture has low inference efficiency



Linear (Recurrent) Attention

- The standard GPT model uses softmax attention
- Linear Attention replaces softmax with a simplified parallel form:

$$O = (QK^\top \odot M)V$$

- It also has an equivalent recurrent form using a state variable S_t:

$$O_t = Q_t S_t, \qquad S_t = S_{t-1} + K_t^T V_t$$

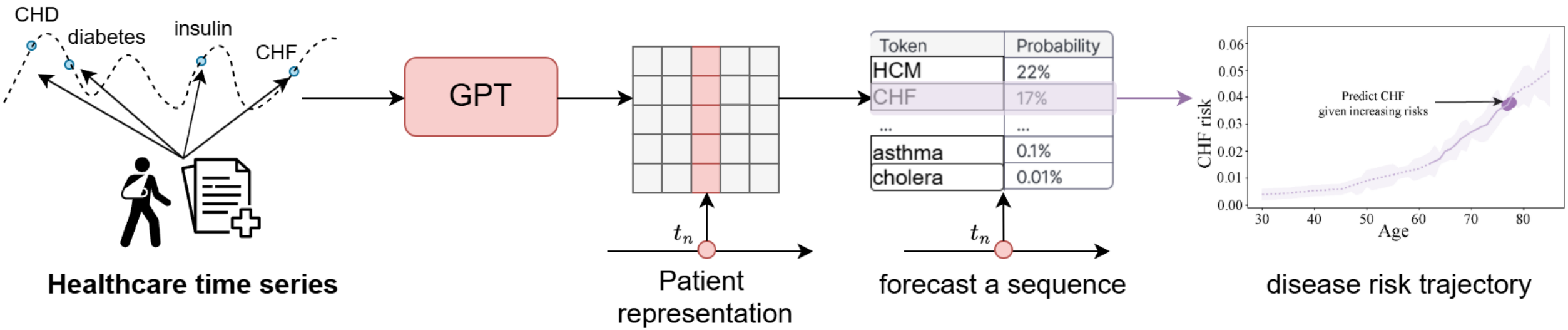
- Linear attention is a recurrent model:
  - It supports the parallel training like GPT
  - It enables efficient inference similar to an RNN

Outline

- Background
- GPT and Recurrent Transformer
- TrajGPT for Trajectory Modeling
- Representation Learning for Medical Concepts
- Reinforcement Learning for LLMs

GPT for Representation learning

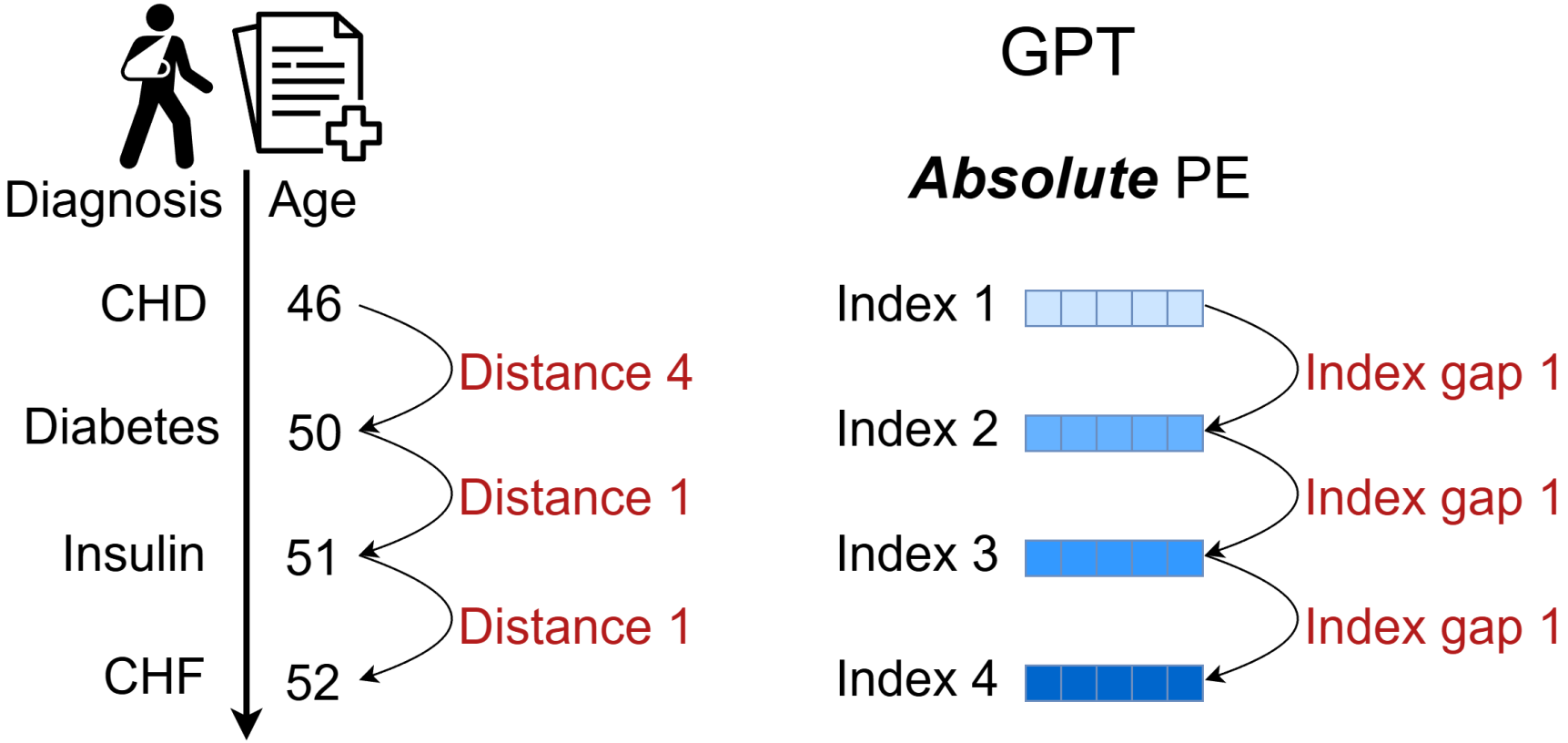
- GPT is a language model (ML) designed for sequence-based representation learning
- Patient representation is utilized to forecast events and generate trajectory



Radford et al. Language Models are Unsupervised Multitask Learners. 2019

Positional Embedding for Varying Time Gaps

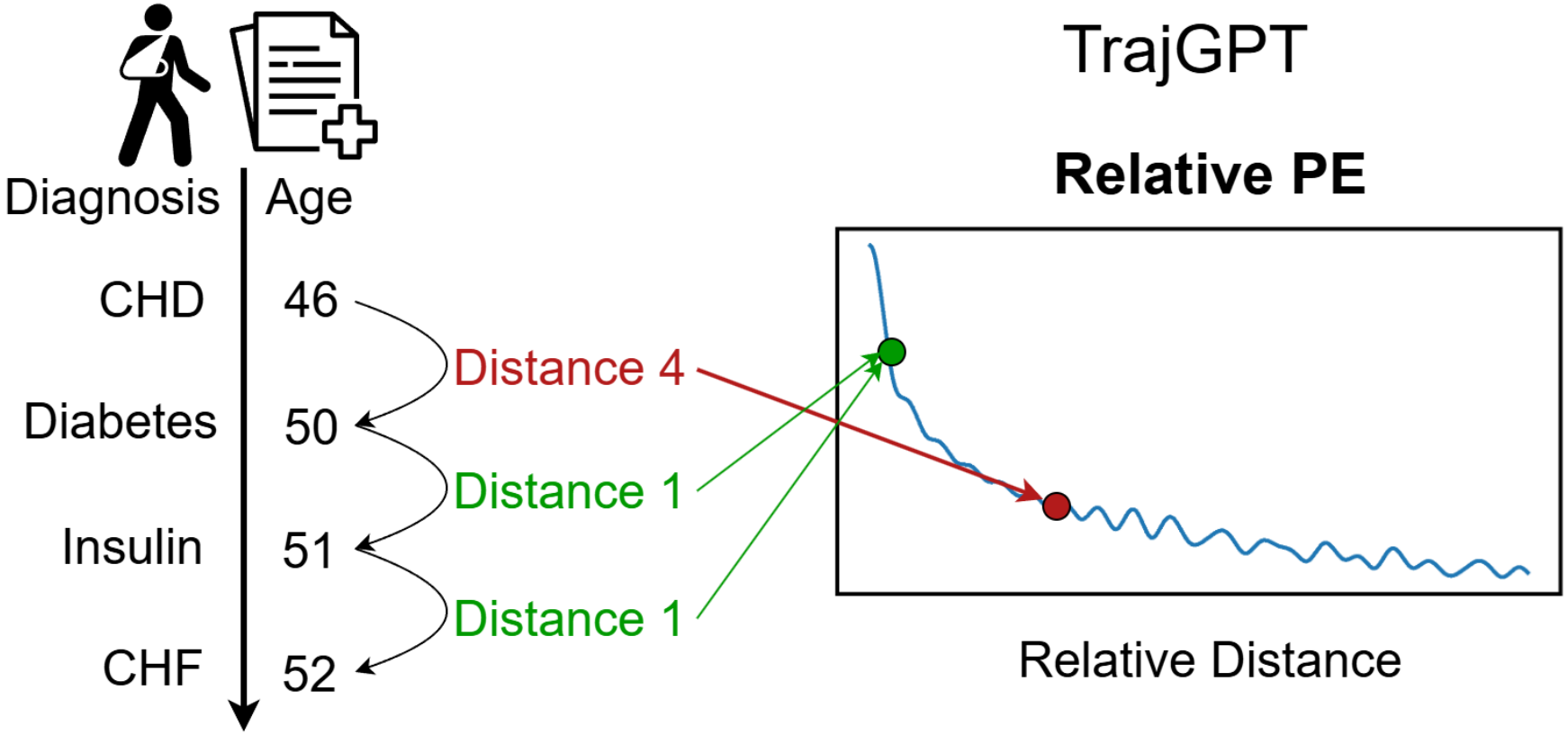
- Absolute positional embedding (PE) only discerns token indices, ignoring varying time interval



Radford et al. Language Models are Unsupervised Multitask Learners. 2019

Relative Position Embedding Handles Irregularity

- Relative PE encodes positional information based on the relative distance of two tokens: age
- Rotary Position embedding (RoPE): $Q_n^T K_m = \left(X_n W_Q\right)^\top \left(e^{i\theta}\right)^{n-m} X_m W_K$



Song et al. Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. ***ACM BCB*** 2024

Relative PE Induces Linear Attention

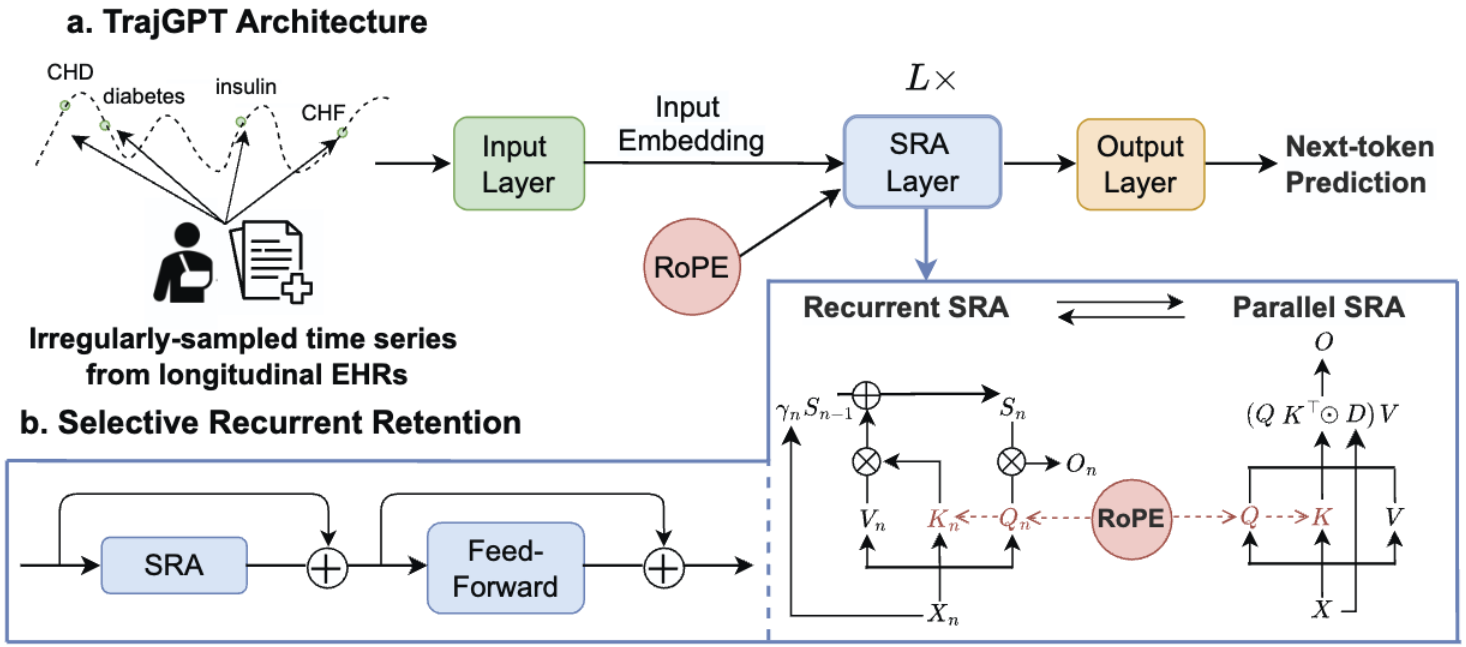
- TrajGPT with RoPE derives a Linear Attention with the following parallel form:

$$Q_n = X_n W_Q e^{i\theta n}, \qquad K_m = X_m W_K e^{-i\theta m}, \qquad V = XW_V$$

$$O = (QK^\top \odot D)V, \qquad D_{nm} = \begin{cases} \gamma^{n-m}, & n \geq m \\ 0, & n < m \end{cases}$$

- It has an equivalent recurrent form of linear attention:

$$O_n = Q_n S_n, \qquad S_n = \gamma^{n-m} S_m + K_n^T V_n$$



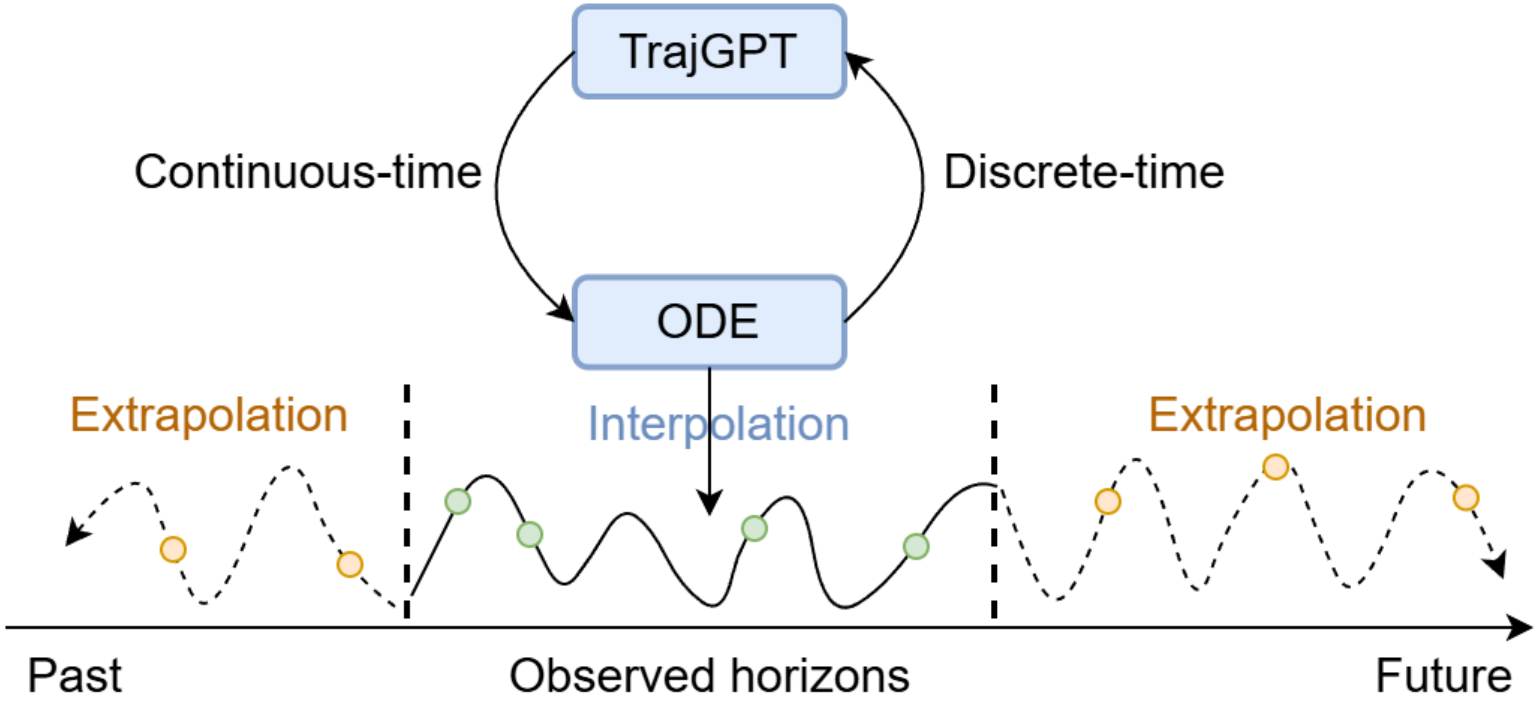
Song et al. TrajGPT: Irregular Time-Series Representation Learning for Health Trajectory Analysis. ***IEEE J-BHI*** 2025.

Linear Attention Is A Discretized ODE

- TrajGPT is an approximated Ordinary Differential Equations (ODE):

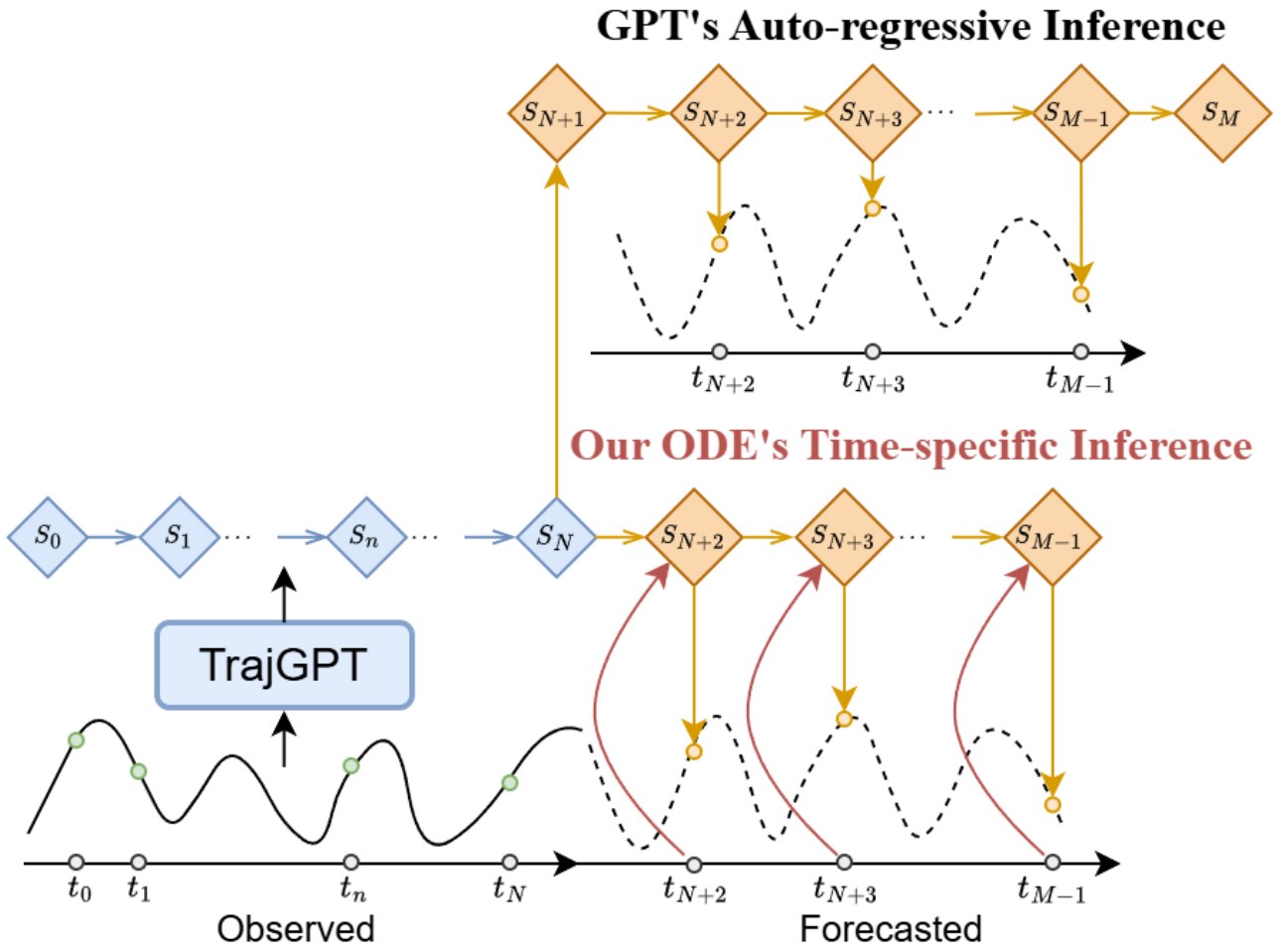
$$S_n = \gamma^{n-m} S_m + K_n^T V_n \approx \int_m^n S'(t)$$

- TrajGPT can **handles irregularity** through exponential decay γ^{n-m}
- It makes interpolation and extrapolation with varying discretize step size $n - m$



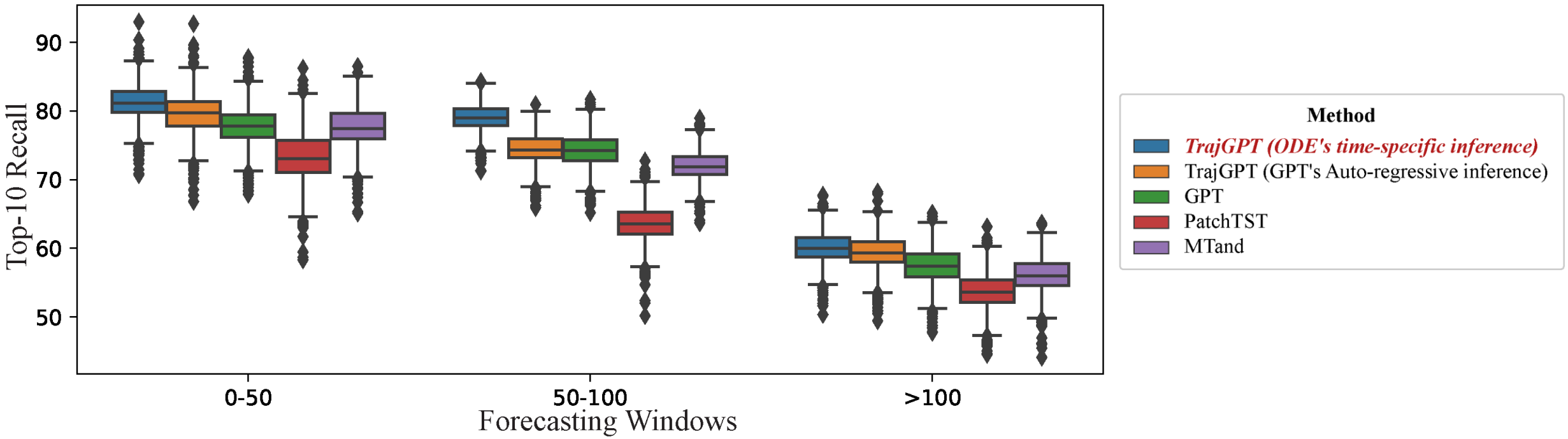
ODE-Inspired Inference Method

- Time-specific inference directly predicts arbitrary target timestep given $n - m$
- Reduce computational steps and error accumulation



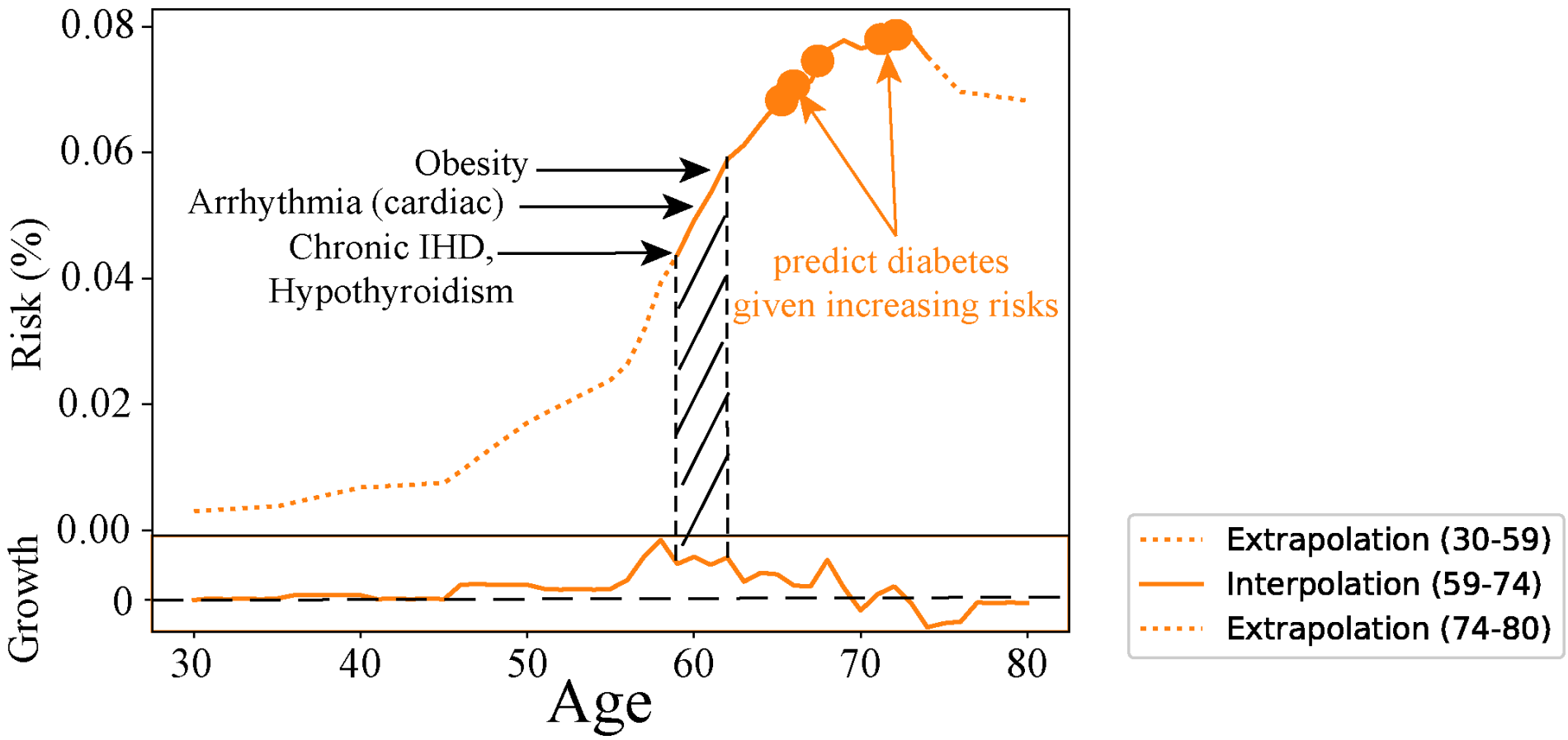
Forecast Irregular-sampled Diagnoses

- We pre-train TrajGPT on the public eICU and private PopHR datasets
- TrajGPT with time-specific inference has more accurate and stable predictions



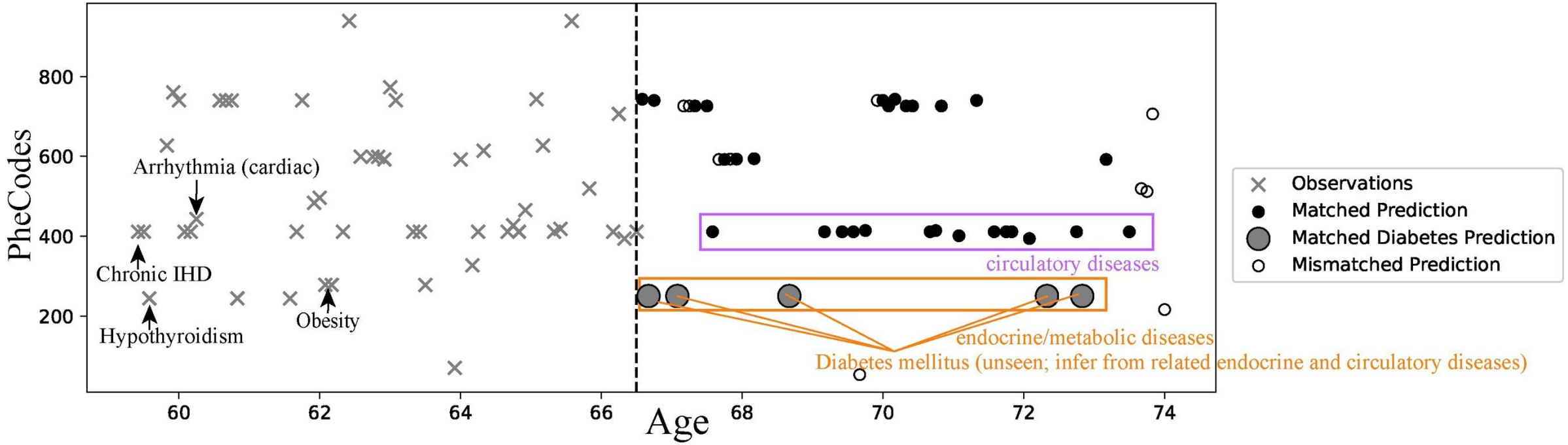
Example Trajectory of A Diabetic Patient

- TrajGPT captures rising diabetic risks and identifies diseases that contribute high risk growth.



Example Trajectory of A Diabetic Patient

- TrajGPT successfully **forecasts unseen disease** beyond the medical history given correlated diseases

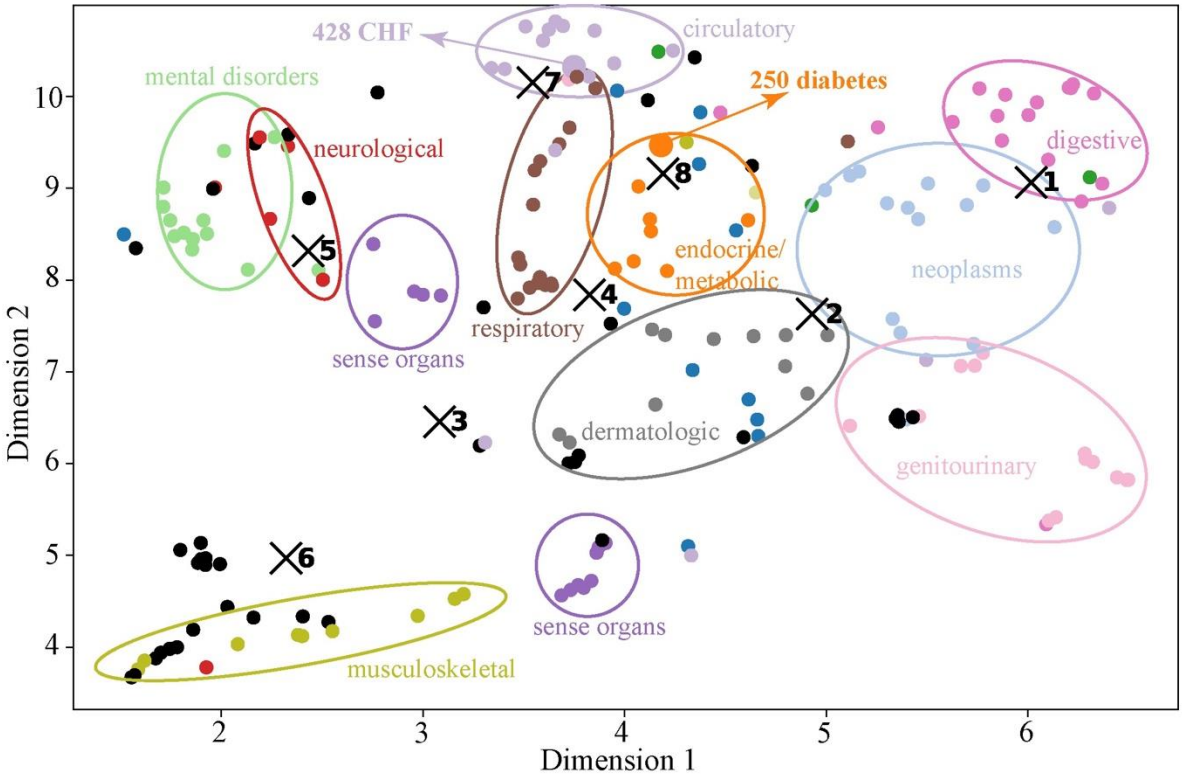


Outline

- Background
- GPT and Recurrent Transformer
- TrajGPT for Trajectory Modeling
- Representation Learning for Medical Concepts
- Reinforcement Learning for LLMs

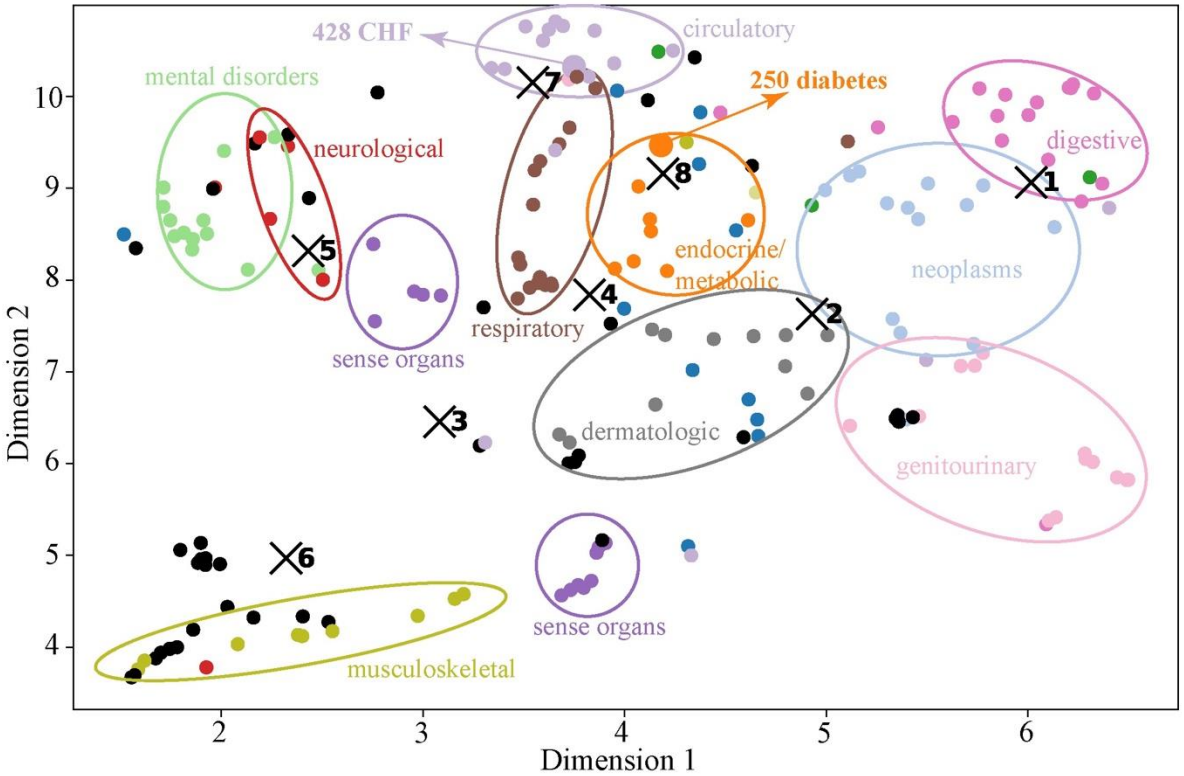
Representation of Diagnostic Codes

- TrajGPT learns high-dimensional embeddings for tokens (diagnosis codes)
- We apply UMAP to project the learned diagnosis embeddings into 2d space
- The learned embedding are clustered with respect to disease categories



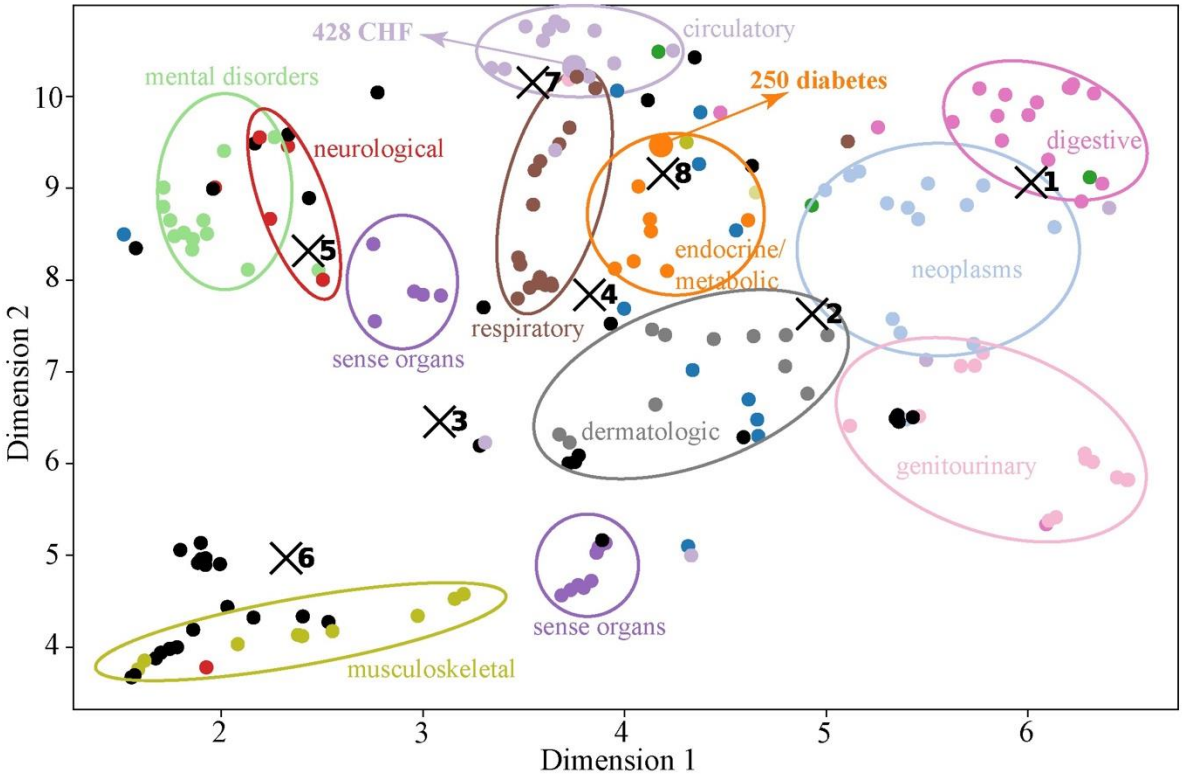
Data Scarcity for Representation Learning

- Limited data makes it challenging to learn representations for fine-grained clinical concepts (specific symptoms).
- We merge diagnosis codes (~10K) to coarse-grained phenotypes (~600).
- We learn embeddings only at the coarse phenotype level with enough data.



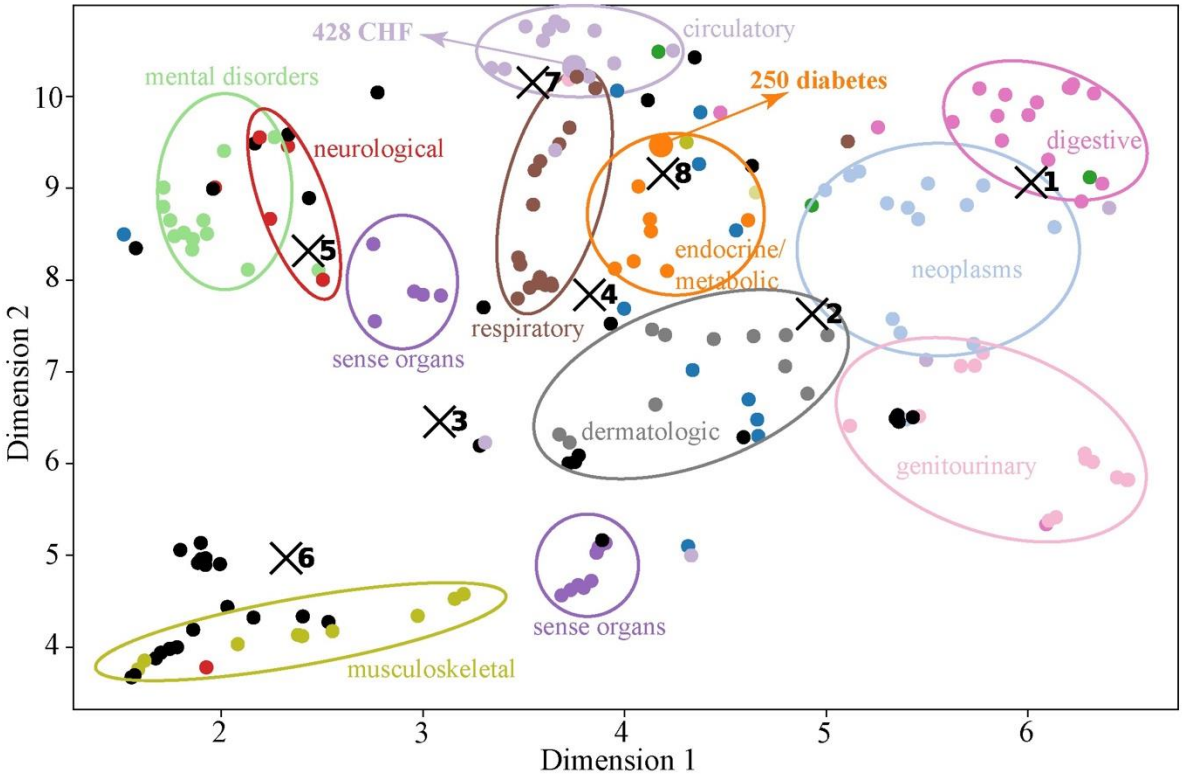
Lack of Explainability in Data-Driven Learning

- Some token embeddings are interpretable:
  - Mental disorder codes cluster near neurological ones
- However, many token embeddings are misaligned.
  - We are unclear whether the model captures meaningful medical relationships or just noise.



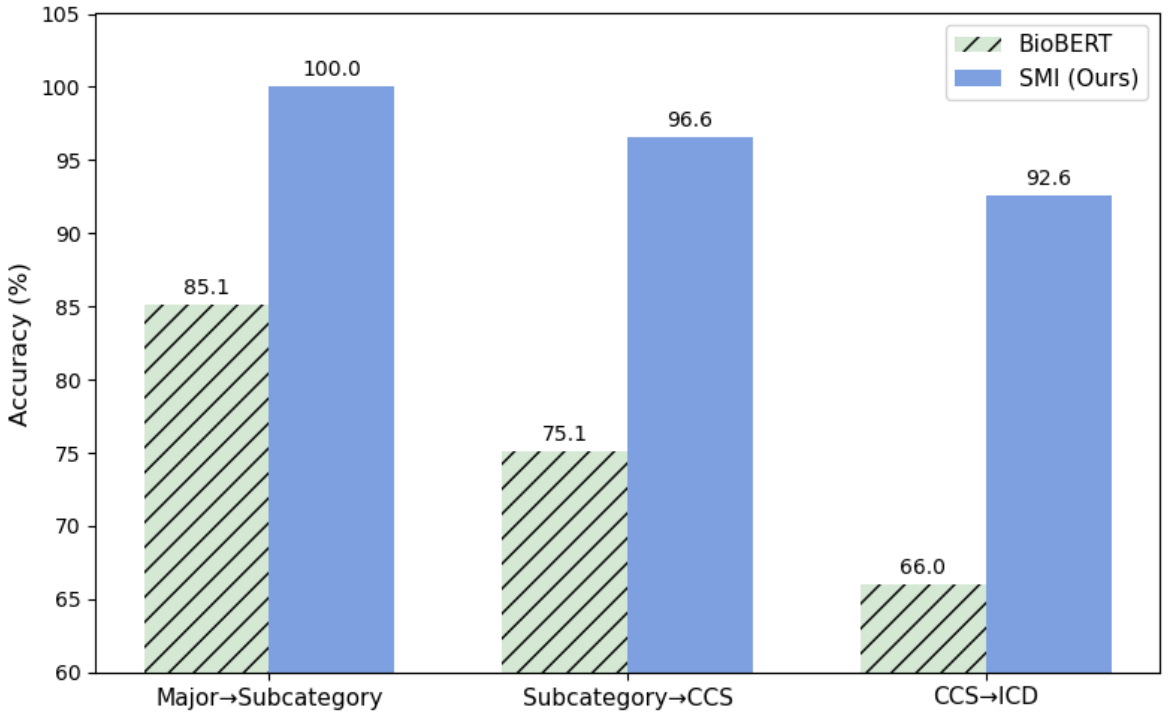
Lack of Explainability in Data-Driven Learning

- Some token embeddings are interpretable:
  - Mental disorder codes cluster near neurological ones
- However, many token embeddings are misaligned.
  - We are unclear whether the model captures meaningful medical relationships or just noise.



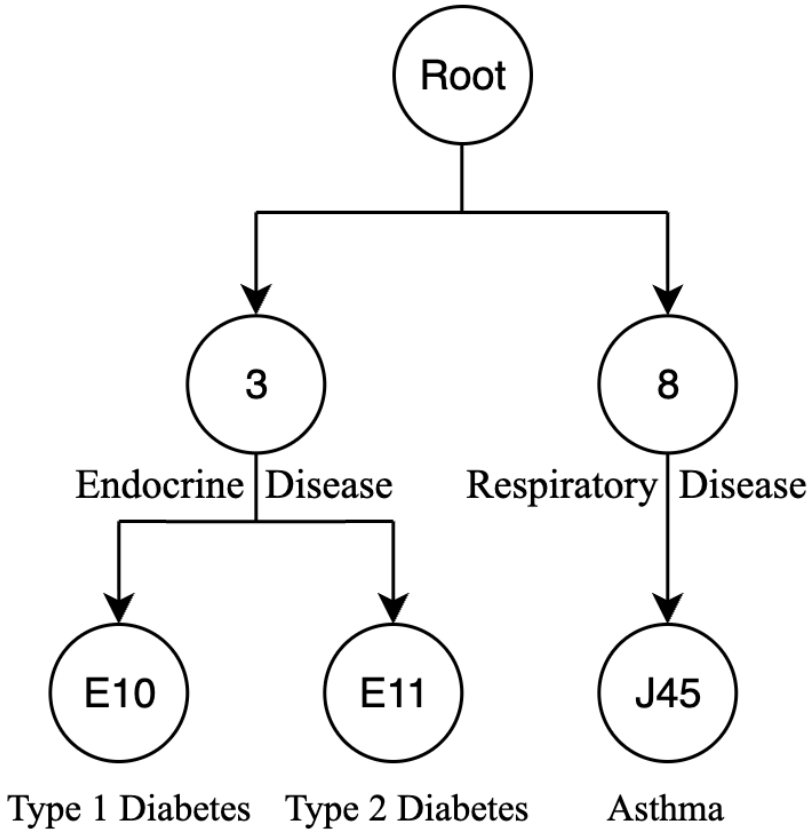
How about Biomedical LMs?

- While Biomedical LMs learn biomedical knowledge from large corpora, but it still have the issue of distinguishing fine-grained concepts.
- We evaluate whether the embedding of a child concept is more similar to its parent than to non-parent concepts across three hierarchical levels.
- Hierarchical similarity accuracy of BioBERT drops as concepts become finer.



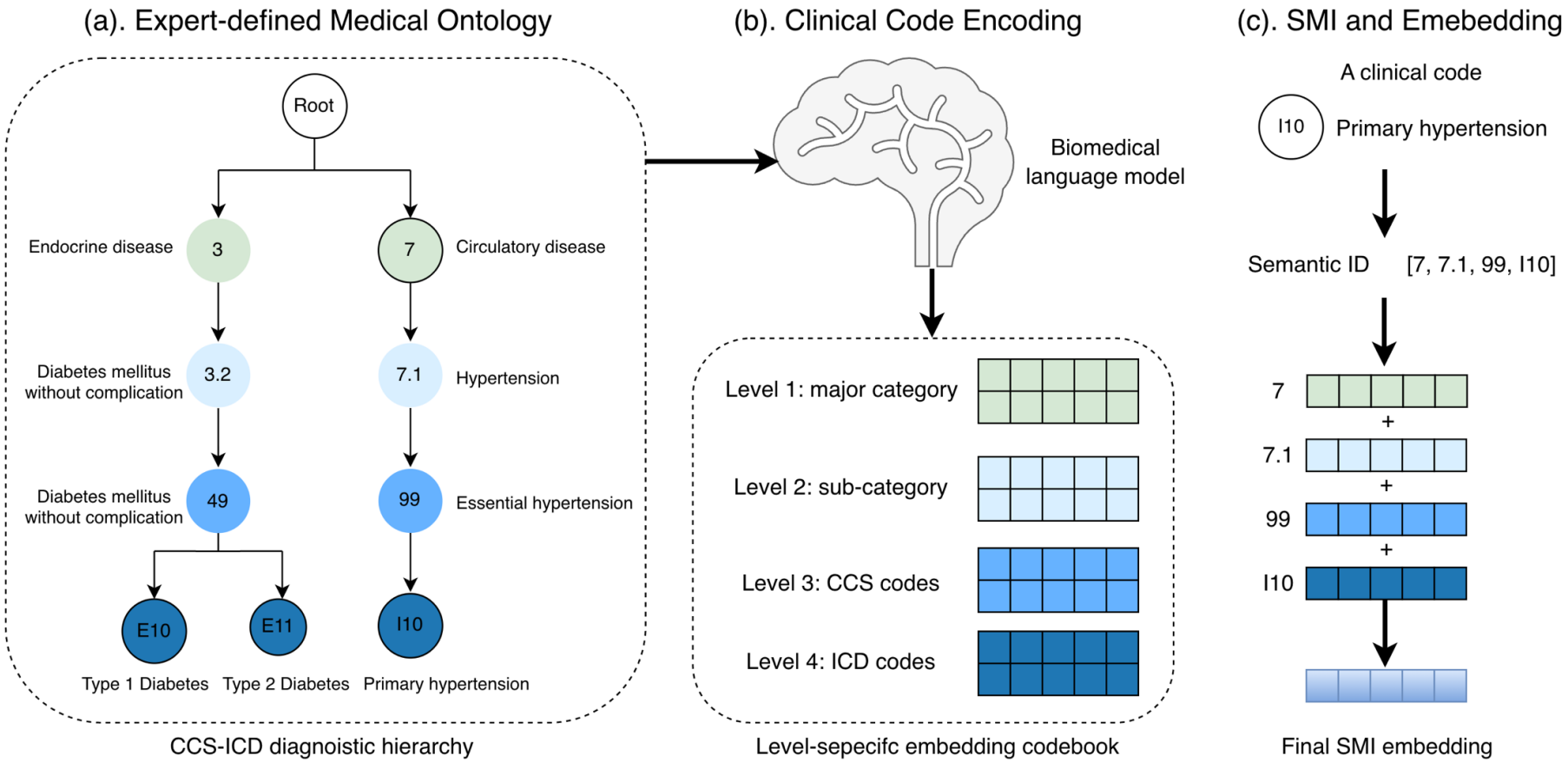
Expert-defined Medical Hierarchy

- We address these two issues by using domain knowledge hierarchy tree, which organizes medical concepts as a tree from broad to specific.
  - Related concepts (e.g.,Type 1 and Type 2 diabetes) are close.
  - Unrelated concepts (e.g.,Type 1 diabetes and Asthma) are distant.



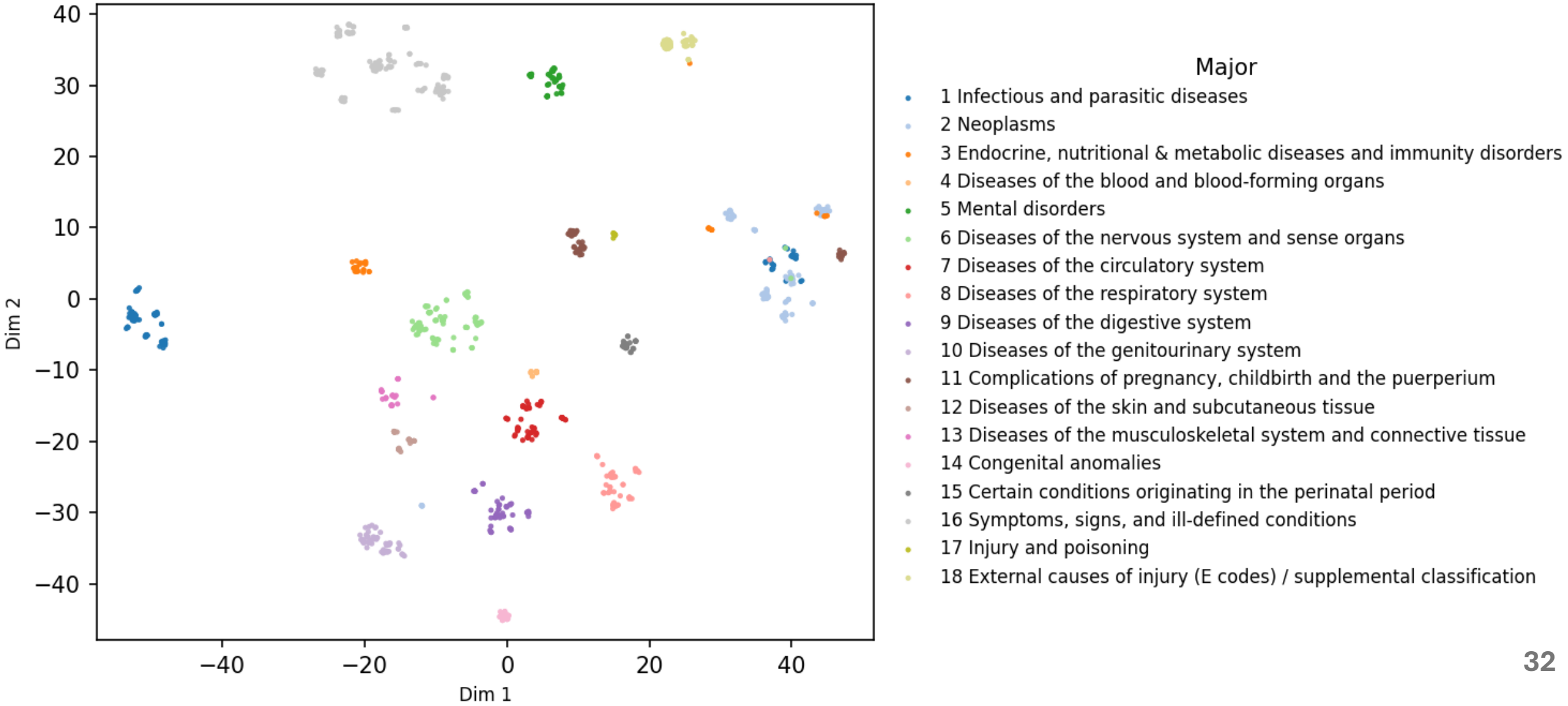
Semantic Medical ID for Embedding

- Given a hierarchy, we use a BioBERT to encode all concepts.
- For each diagnosis, we extract its related concepts at different levels to build a semantic ID, and sum over embeddings over different levels to obtain final embedding.



Visualization of SMI's Embedding

- We use TSNE with default setup to visualize 768d SMI's embedding.



Visualization of SMI's Hierarchy

- We compare the learned hierarchy with the expert-defined hierarchy.

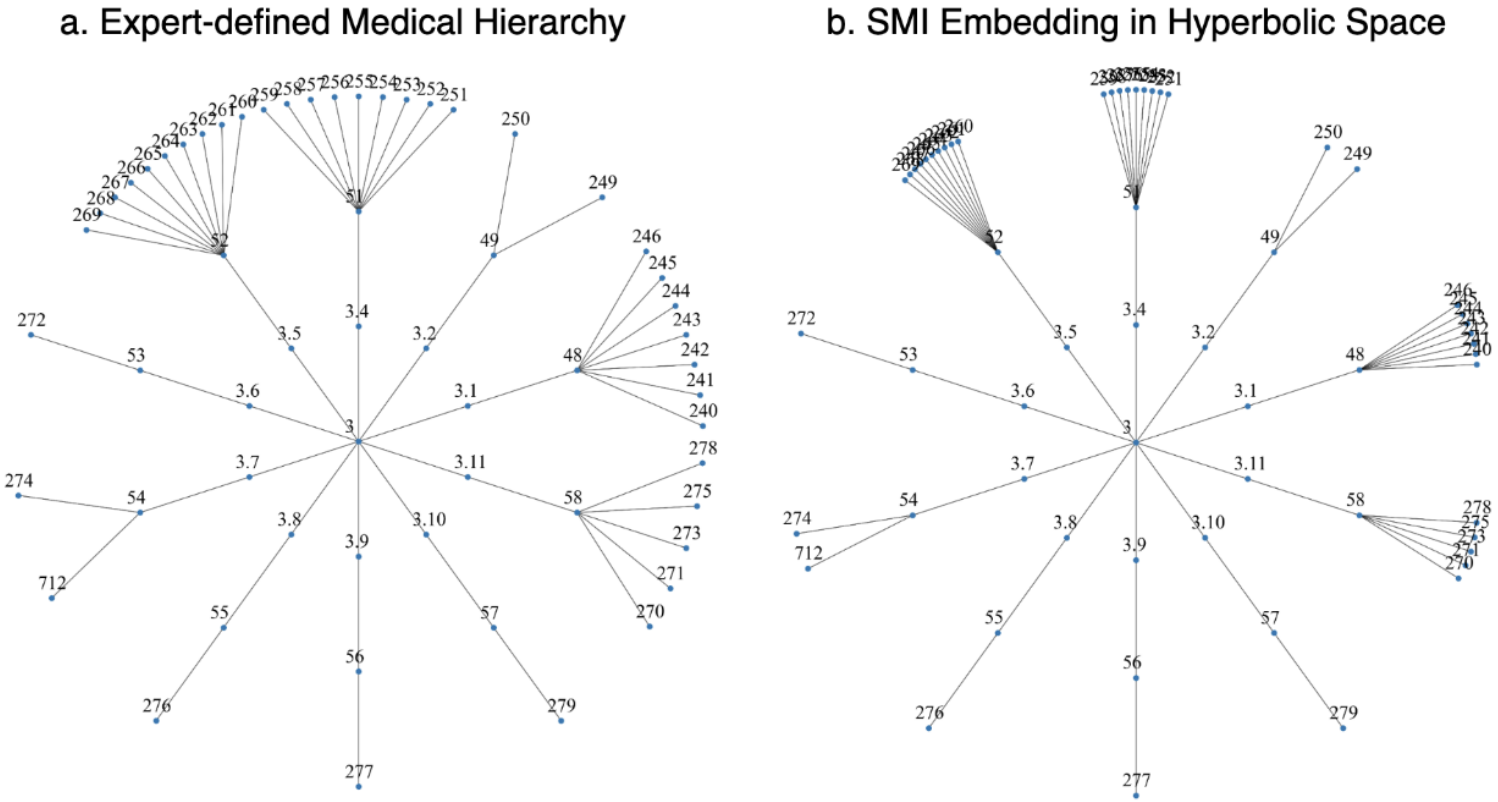


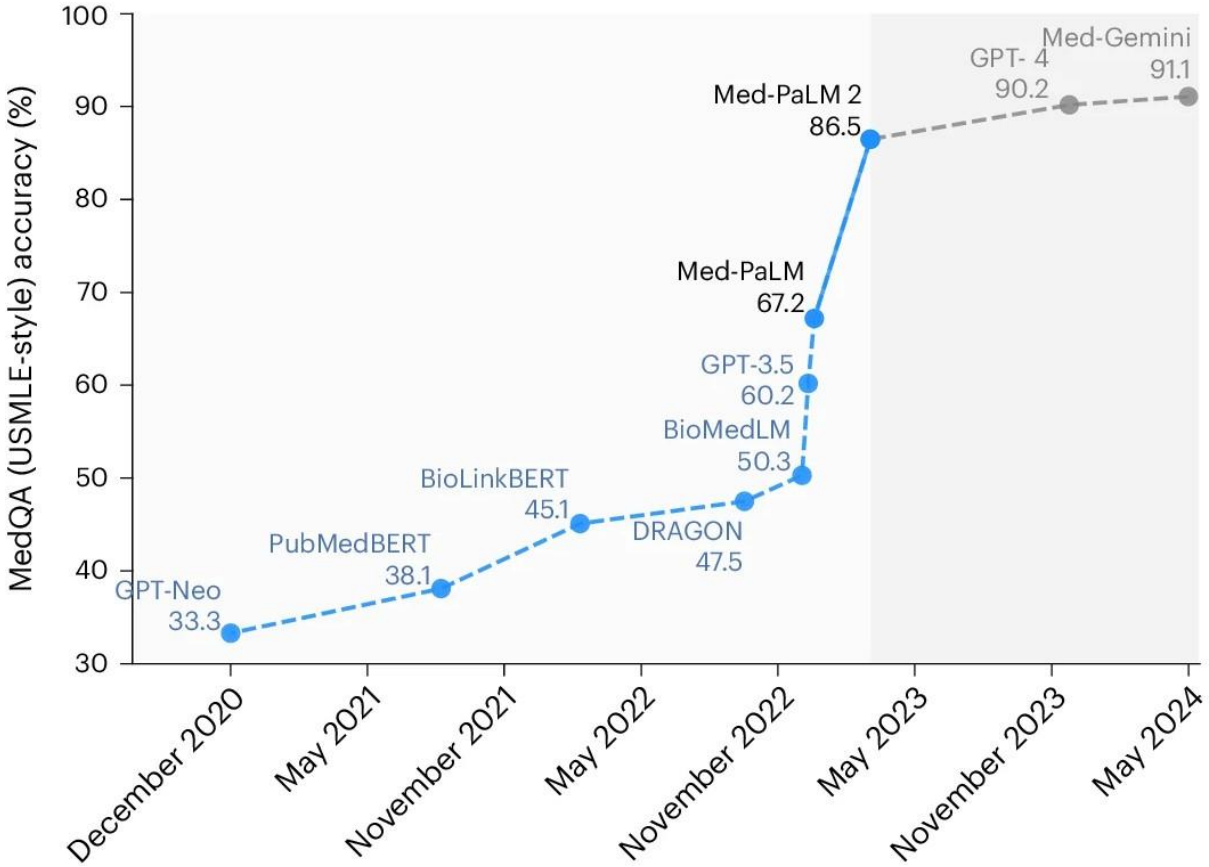
Figure 4: Visualization of **Major Disease Category 3: Endocrine, Nutritional, and Metabolic Diseases and Immunity Disorders** both the expert-defined medical ontology and in SMI embeddings within hyperbolic space. (a) Expert-defined ICD hierarchy illustrates all descendent concepts under circulatory system diseases. (b) SMI embeddings of the same concepts, projected onto the Poincaré ball to visualize pairwise relationships in hyperbolic distance. The illustrations demonstrate that the SMI embedding inherently learns representations in hyperbolic geometry that naturally reflect the hierarchical structure of the expert-defined medical ontology.

Outline

- Background
- GPT and Recurrent Transformer
- TrajGPT for Trajectory Modeling
- Representation Learning for Medical Concepts
- Reinforcement Learning for LLMs

AI4health Research in the Age of LLMs

- Biomedical LLMs show an improved understanding of medical knowledge
- Apply biomedical LLMs to healthcare by ensuring explainability?

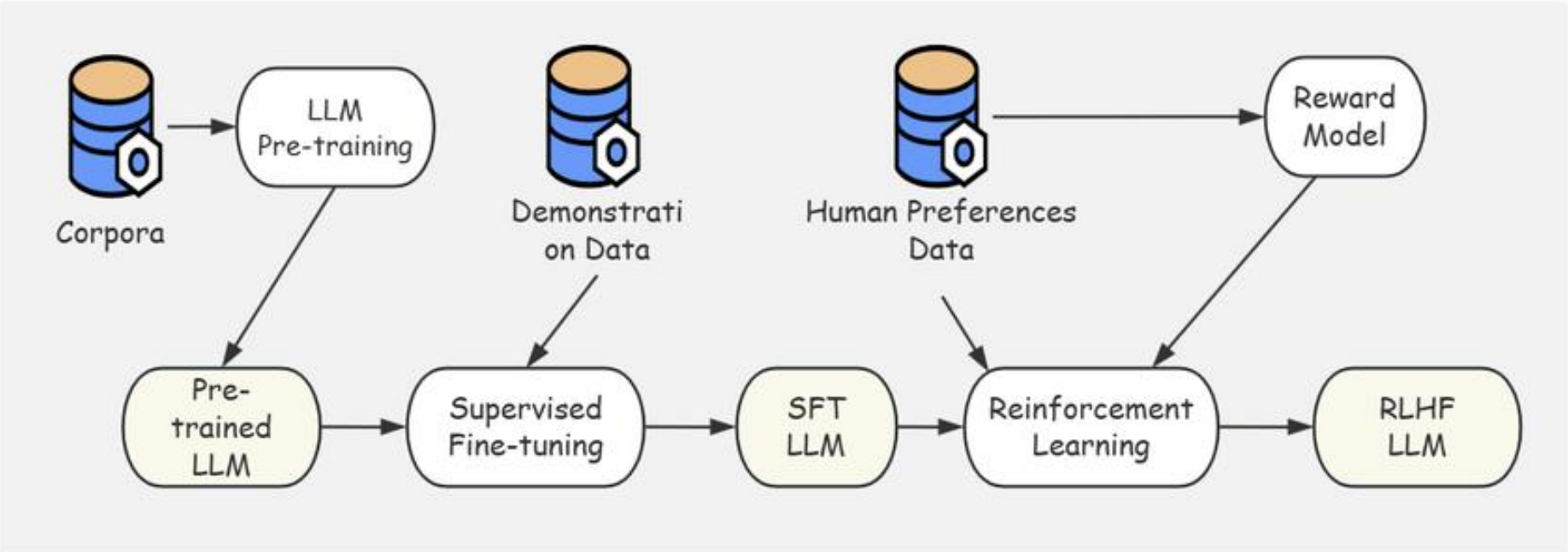


Large language models encode clinical knowledge. ***Nature*** 2023

Toward expert-level medical question answering with large language models. ***Nature Medicine*** 2025

Reinforcement Learning in LLMs

- LLMs are first pre-trained on large-scale corpora to acquire knowledge.
- Supervised Fine-Tuning (SFT) adapts the model to human-written examples for specific tasks.
- Reinforcement Learning(RL) further aligns the LLM outputs with human preference, known as Reinforcement Learning from Human Feedback (RLHF).



The DeepSeek Moment for Reasoning

- DeepSeek shows rules boost AI reasoning, even for logic-heavy math.
- DeepSeek uses RL with rule-based rewards to align outputs with expert knowledge.
- Same recipe can strengthen biomedical LLM reasoning using expert-defined medical rules.

deepseek

Article | Open access | Published: 17 September 2025

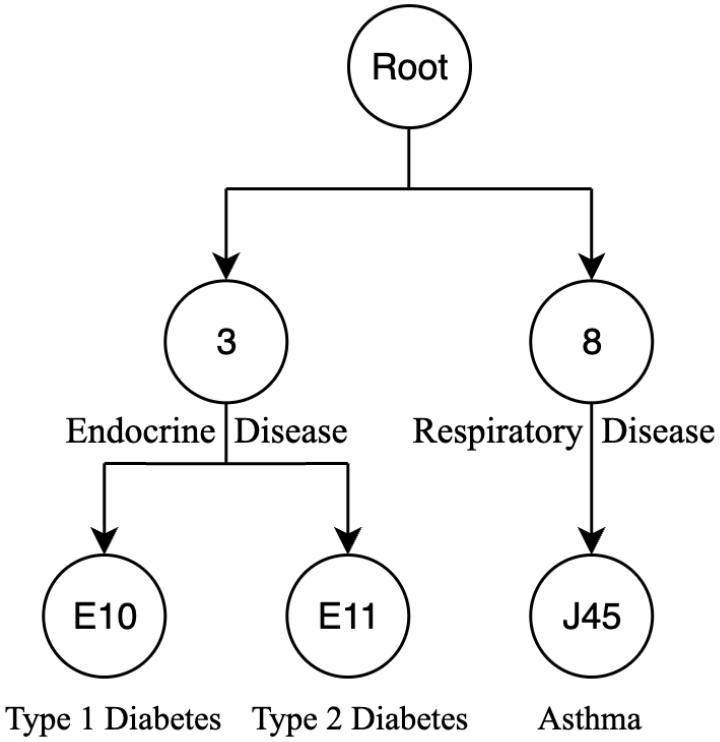
DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning

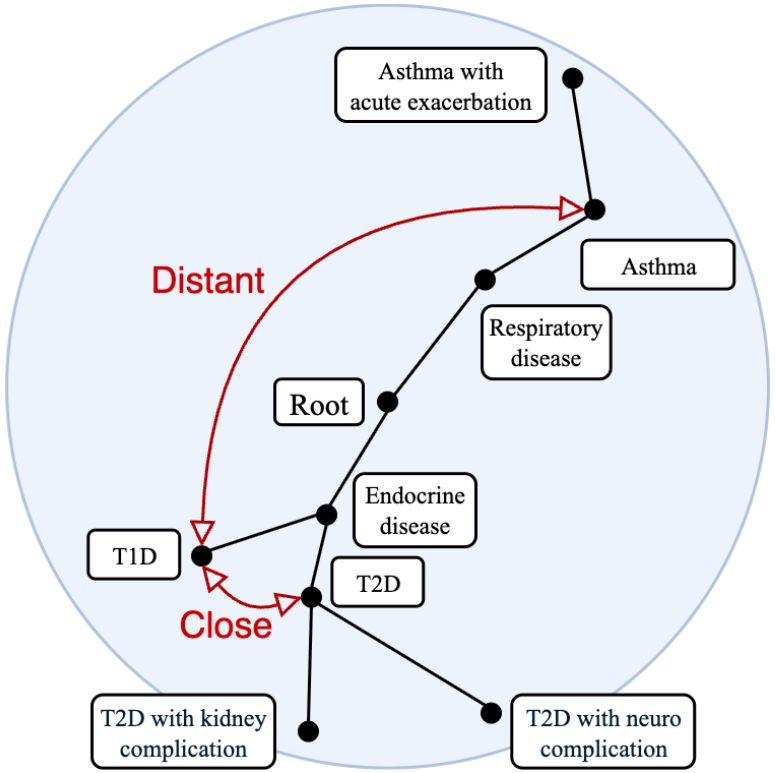
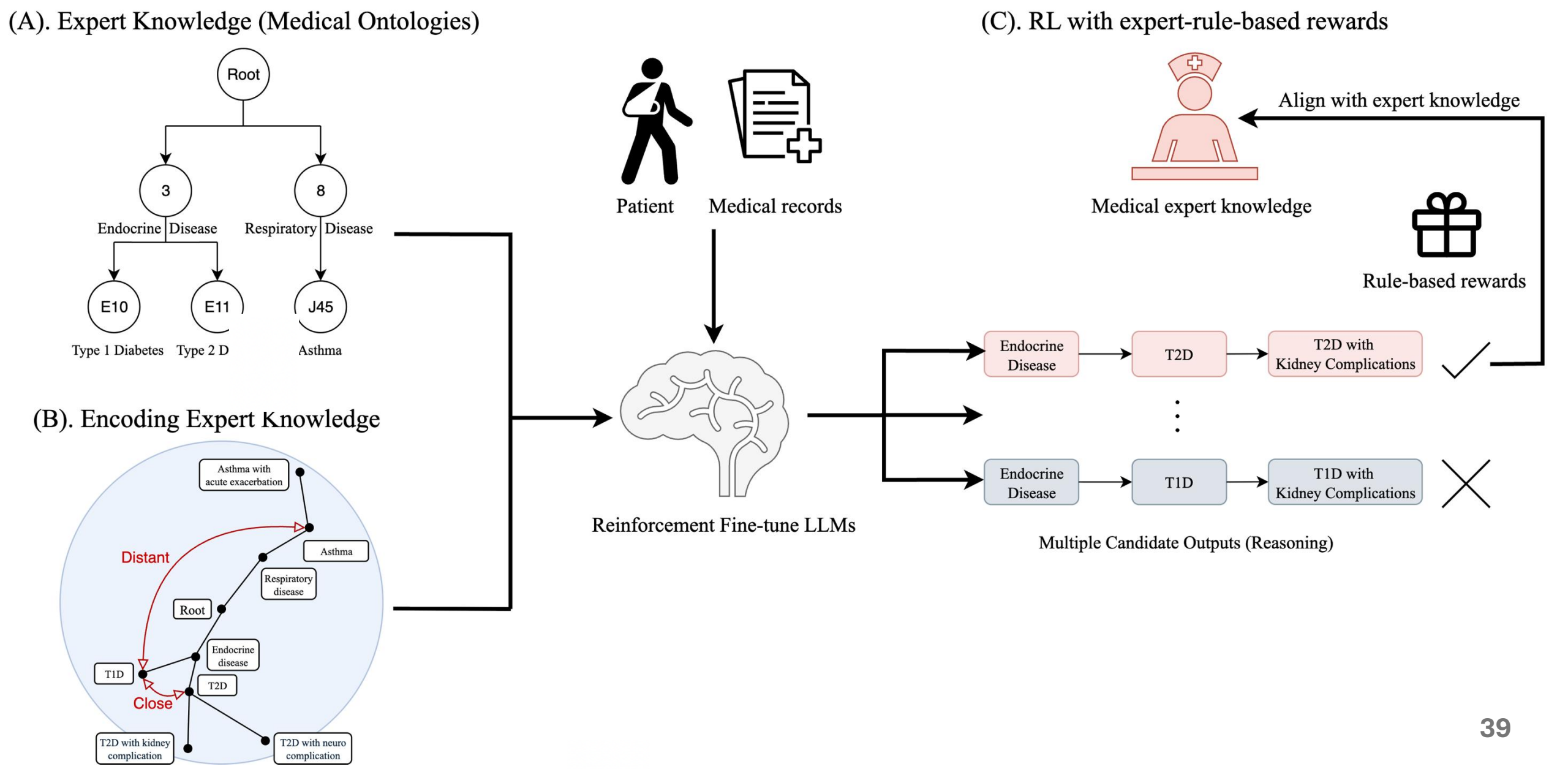
Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, ... Zhen Zhang + Show authors

Nature **645**, 633–638 (2025) | Cite this article

Medical Hierarchy Is Expert-defined Reasoning

- The medical concepts are learned in a tree-based representation space.
- It enables step-by-step reasoning (e.g., Endocrine disease →Type 2 diabetes →Type 2 diabetes with kidney complications).
- We want the LLM outputs as close as possible to expert-defined hierarchy.





Align LLMs with Medical Rules